

Assessify: Comprehensive UI, UX & Business Process Analysis

Teacher English Placement Testing Engine · Yayasan Karya Bangsa (TK, SD, SMP, SMA, SMK)

Evaluation Standards: CEFR (A1–C1) & IELTS Descriptors | Platform Version: 2026.3 | Status: Verified Live

1. Executive Overview & Organizational Mission

Assessify is a specialized, institutional-grade placement assessment platform engineered for Yayasan Karya Bangsa. The system provides a standardized, objective, and auditable methodology for benchmarking the English language competencies of all instructional staff across the foundation's 5 educational units: TK, SD, SMP, SMA, and SMK Karya Bangsa. By establishing an empirically grounded baseline aligned with the Common European Framework of Reference for Languages (CEFR: A1, A2, B1, B2, C1), the institution ensures pedagogical excellence and targeted professional development programs.

Core Strategic Capabilities:

Standardized Multi-Skill Testing

Assesses Listening (15 items), Grammar & Vocabulary (20 items), Reading (15 items), Writing (2 items), and Speaking (2 items) in a controlled 75-minute assessment window.

Dual-Layer Evaluation Engine

Instant objective answer-key scoring for receptive skills combined with an administrative 4-criteria manual evaluation matrix (1–5 scale) for productive skills (Writing & Speaking).

Hardware Diagnostics & Proctoring

Integrated browser pre-flight checks validating audio playback, microphone level detection via Web Audio API, and camera stream capture via MediaRecorder.

Multi-Unit Governance (TK, SD, SMP, SMA, SMK)

Full school unit segregation, single & bulk PDF certificate generation, formatted Excel (.xlsx) export, and in-platform asynchronous speaking audio/video evaluation.

2. Key System Stakeholders & Role Matrix

Teacher Candidates (Examinees)

- Secure sign-in via school domain email (@karyabangsa.sch.id) & unit selector (TK/SD/SMP/SMA/SMK).
- Interactive 5-stage assessment with persistent countdown timer & auto-saving.
- Audio playback with text-to-speech for Listening & video recording for Speaking.
- Instant receipt of provisional placement estimate upon assessment completion.

School Administrators & Evaluators

- Role-protected administrative dashboard with password hashing.
- Multi-unit candidate filtering (TK, SD, SMP, SMA, SMK), candidate search, and bulk deletion.
- Interactive rubric grading modal (4 criteria, 1–5 scale) with real-time CEFR band logic.
- Single & bulk PDF placement certificate generator and filtered Excel export.

3. Complete Business Process Architecture & Workflow

The operational lifecycle in Assessify encompasses four interconnected workflow phases: examinee authentication & readiness checks, live assessment execution, administrative rubric grading, and credential generation & reporting.

1	Phase 1: Candidate Verification & Diagnostics Teacher inputs name, school-domain email, and selects school unit (TK, SD, SMP, SMA, or SMK Karya Bangsa). The system runs an automated client-side diagnostic: audio speaker test, microphone input level meter, camera feed validation, and network stability check before issuing assessment access.
2	Phase 2: 75-Minute Proctored Assessment Examinee completes 5 sequential sections: (1) Listening Comprehension (TTS audio prompts), (2) Grammar & Vocabulary (contextual multiple choice), (3) Reading Comprehension (passage analysis), (4) Writing Essay (rich text editor with live word count), and (5) Speaking Interview (in-platform video/audio recording). Server tracks attempt timer.
3	Phase 3: Automated Scoring & Ingestion Objective sections (50 items total) are evaluated immediately against JSON answer keys and mapped to CEFR bands (A1–C1). Productive sections (Writing & Speaking) are queued with recording media for administrative review. Provisional overall estimate is computed.
4	Phase 4: Admin Rubric Review & Governance Evaluator reviews candidate essay and plays recorded speaking video directly inside the admin console. Admin scores 4 criteria for Writing and 4 criteria for Speaking on a 1–5 scale. The system calculates section totals, assigns final CEFR bands, and issues official certificates.

4. Integration & Infrastructure Touchpoints

In-Platform Media Capture & Asynchronous Rubric Review Engine Assessify operates a self-contained asynchronous oral assessment pipeline. Examinees record their speaking responses directly inside the browser using HTML5 MediaRecorder and Web Audio API visualizers. Recordings are uploaded with HTTP 206 Partial Content streaming support, allowing evaluators to audit recordings and score rubrics directly in-app with zero third-party meeting dependencies.	
Storage & Persistence Layer • Primary: MySQL database with InnoDB engine. • High-performance indices on email, unit, status. • Automatic in-memory development fallback. • Local media storage with HTTP 206 partial streaming.	Reporting & Export Services • ExcelJS: Multi-worksheet data exports with styling. • PDFKit: Vector-rendered placement certificates. • Real-time candidate search & 5-unit filtering. • JSON Question Bank and Rubrics schema validation.

5. Design System, Visual Hierarchy & Color Psychology

Assessify utilizes a modern, institutional-grade design system crafted in clean Vanilla CSS. The visual hierarchy emphasizes cognitive clarity, low examinee anxiety during timed assessments, and high information density for administrative dashboards across all 5 school units.

CEFR Placement Level Color Palette:

A1 · Beginner	A2 · Elementary	B1 · Intermediate	B2 · Upper-Intermediate	C1 · Advanced
Crimson Red Basic formulaic expressions; limited syntactic control.	Warm Amber Routine social exchange; simple grammatical structures.	Royal Blue Independent communication; main ideas of familiar matters.	Emerald Green Fluent academic interaction; complex argument synthesis.	Royal Purple Full professional mastery; sophisticated idiom & flexible syntax.

6. Key UI/UX Pillars & Interaction Patterns

Progressive Disclosure & Navigation

The 5 test sections are presented sequentially with clear progress indicators, section timers, and forward/backward navigation controls. Candidates cannot accidentally jump ahead without completing necessary prerequisites.

Accessible Toast Notification Architecture

Replaces disruptive native browser alerts with custom floating toast notifications (Success, Error, Info) rendered with clear SVG iconography, auto-dismiss timers, and manual close triggers.

Micro-Interactions & Real-Time Feedback

Live Web Audio frequency visualizer reflects voice input levels during speaking checks; active word counter dynamically updates during essay writing; rubric buttons toggle instantly with immediate score computation.

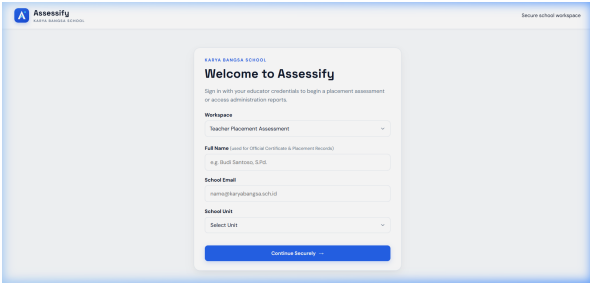
High-Contrast Typography & Visual Focus

Standardized system font stack (Inter, system-ui, -apple-system) with rigorous contrast ratios (WCAG AAA compliant), focused card containers, and distinct active state outlines.

7. Examinee Onboarding & Diagnostic Pre-Test Flow

Before starting the timed assessment, teachers complete authentication and an automated hardware diagnostics suite. This guarantees optimal test conditions and prevents technical failures during the timed speaking or listening sections.

LIVE APPLICATION CAPTURE: Teacher Authentication & School Unit Selection Modal (TK, SD, SMP, SMA, SMK)



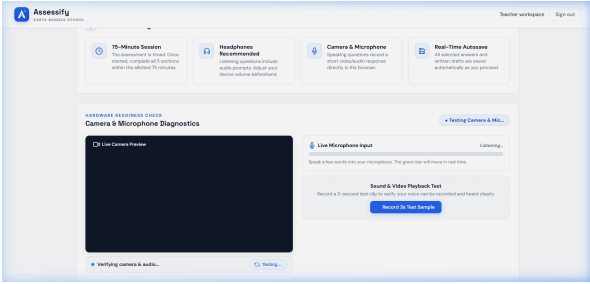
UI & UX Design Strengths

- Domain enforcement (@karyabangsa.sch.id) ensures institutional data integrity.
- Clean role switcher toggle between Teacher Assessment and Admin Workspace.
- Mandatory School Unit dropdown (TK, SD, SMP, SMA, SMK) prevents unassigned records.

Technical & Architectural Rigor

- HMAC-SHA256 encrypted cookie sessions (`assessify\_session`) with Lax SameSite.
- Full Name auto-inferred from email prefix when omitted.
- Immediate client-side validation prevents invalid form submission.

LIVE APPLICATION CAPTURE: Hardware Diagnostics Suite (Speaker, Microphone Meter, Camera & Rules)



Diagnostic Pre-Flight Capabilities

- Interactive audio test button plays an embedded Web Audio chime.
- Real-time microphone level meter detects vocal input.
- Clear 75-minute assessment rules and integrity guidelines.

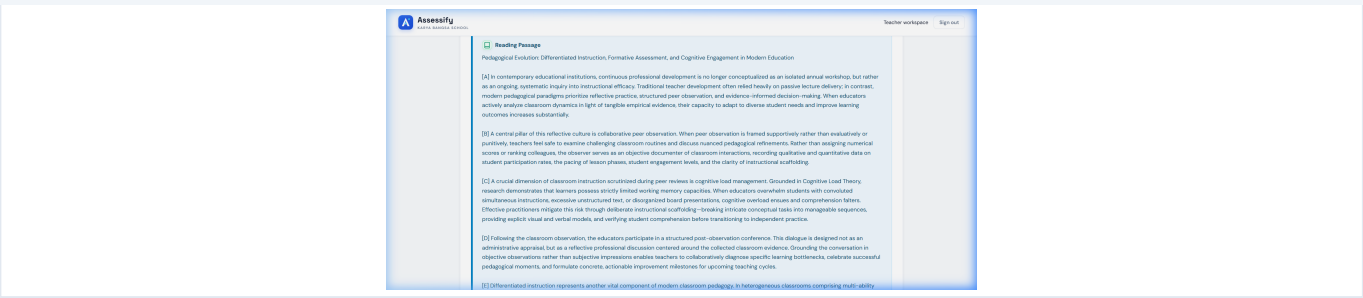
Failure Recovery & Error Handling

- Explicit permission retry prompts for camera/mic access.
- Warning dialogs if hardware is disconnected.
- "Begin Assessment" button enables only upon user confirmation.

8. Receptive & Written Production Assessment Modules

Assessify presents questions in focused cards with distinct typography. Reading passages utilize a side-by-side or scroll-synchronized layout, and the Writing section provides an ergonomic essay editor.

LIVE APPLICATION CAPTURE: Reading Comprehension Module with Passage Context & Objective Items



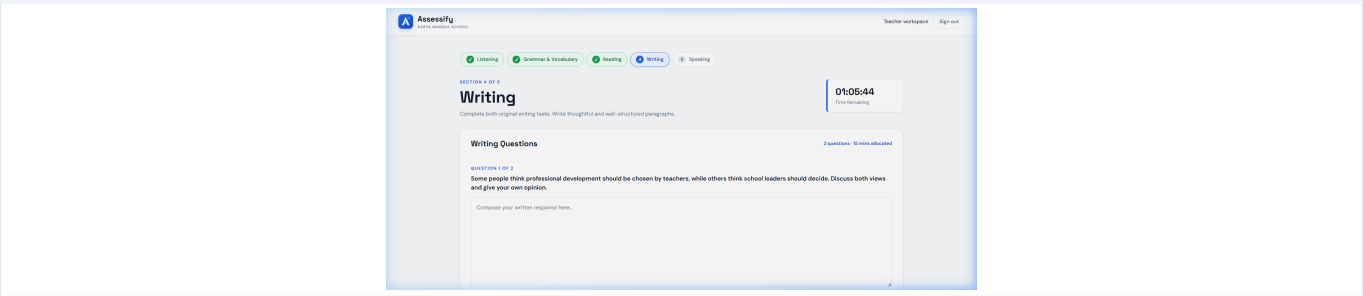
Reading UI/UX Architecture

- Passage is pinned with clear paragraph numbering and academic formatting.
- Multiple choice options feature generous click targets and hover states.
- Selected answers display distinct filled radio indicators.

Grammar & Vocabulary Design

- Contextual sentence-completion items with highlighted target blanks.
- Immediate local response caching prevents loss on accidental page reload.
- Clean item numbering with instant section jump links.

LIVE APPLICATION CAPTURE: Writing Assessment Module with Prompt Descriptors & Live Word Count



Writing Interface Ergonomics

- High-contrast textarea with auto-expanding line height.
- Real-time word counter with recommended threshold guidance (150–250 words).
- Automatic draft storage in browser LocalStorage alongside periodic server autosave.

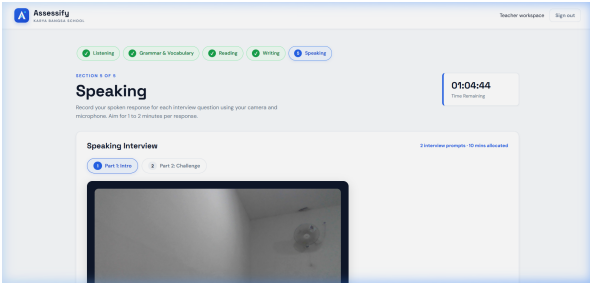
Assessment Integrity Measures

- Paste event logging and minimum engagement timers.
- Clear task achievement descriptors highlighting prompt requirements.
- Warning toast if candidate attempts section advance with empty response.

9. Speaking Module & Assessment Submission

The Speaking module integrates browser media recording with a visual feedback loop. Upon completion, the candidate receives an immediate provisional score summary and official submission confirmation.

LIVE APPLICATION CAPTURE: Speaking Assessment Interface (Live Camera Stream, Mic Visualizer & Recorder)

The screenshot shows the 'Speaking' assessment interface. At the top, there's a progress bar with five items: Listening, Grammar & Vocabulary, Reading, Writing, and Speaking (which is active). Below this, it says 'SECTION 5 OF 6' and 'Speaking'. A timer on the right shows '01:04:44' with 'Time Remaining'. The main area is titled 'Speaking Interview' and has two tabs: 'Part 1: Intro' (active) and 'Part 2: Challenge'. Below the tabs is a live video stream of a person speaking. To the right of the video is a microphone visualizer and a recording button.

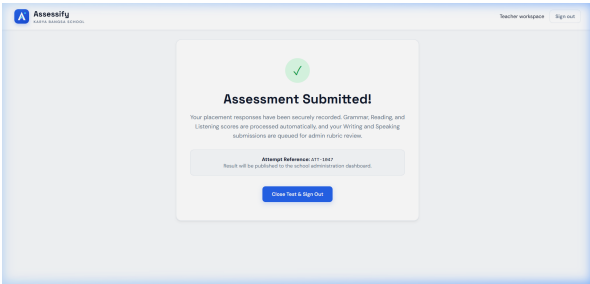
Speaking Interface Capabilities

- Live video stream preview with audio wave meter displaying microphone gain.
- Clear Start/Stop recording toggle with active recording duration timer.
- Fallback audio-only mode for devices without functional webcams.

Media Processing & Security

- Browser MediaRecorder captures high-quality WebM/MP4 media chunks.
- Streaming upload to `/api/attempt/:id/recording` with HTTP range support.
- Maximum file size safeguards (14MB) with client-side compression checks.

LIVE APPLICATION CAPTURE: Assessment Completed Screen with Provisional Placement & Attempt Summary

The screenshot shows the 'Assessment Submitted!' screen. It features a large green checkmark at the top. Below it, the text 'Assessment Submitted!' is followed by a paragraph explaining that responses have been recorded and scores are being processed. An 'Attempt Reference: 011-1847' is displayed, along with a note that the result will be published to the school administration dashboard. At the bottom, there is a 'Close Test & Sign Out' button.

Results Screen UX Highlights

- Instant provisional CEFR placement badge with skill breakdown.
- Reassurance notice explaining that Writing & Speaking undergo teacher review.
- Clear Attempt ID reference and timestamp for administrative correspondence.

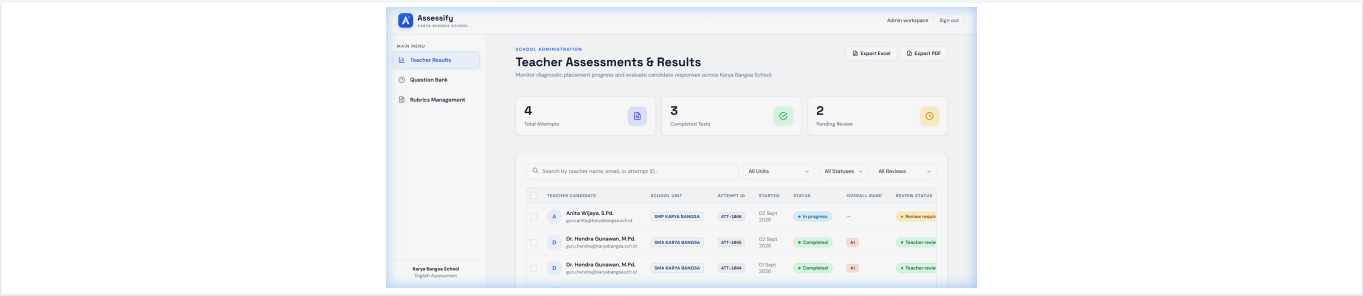
Candidate Post-Submission State

- Candidate attempt status transitions to "Completed" (locked against edits).
- Video/audio media is safely attached to the attempt for administrator grading.
- Secure sign-out trigger clearing local session tokens.

10. Administrative Results Console & Multi-Unit Governance

The Administrator Workspace gives school leadership full visibility across all teacher assessments, enabling fast candidate search, 5-unit filtering (TK, SD, SMP, SMA, SMK), single & bulk data exports, and batch management.

LIVE APPLICATION CAPTURE: Admin Results Console with Real-Time Search, Unit Filtering & Candidate Table



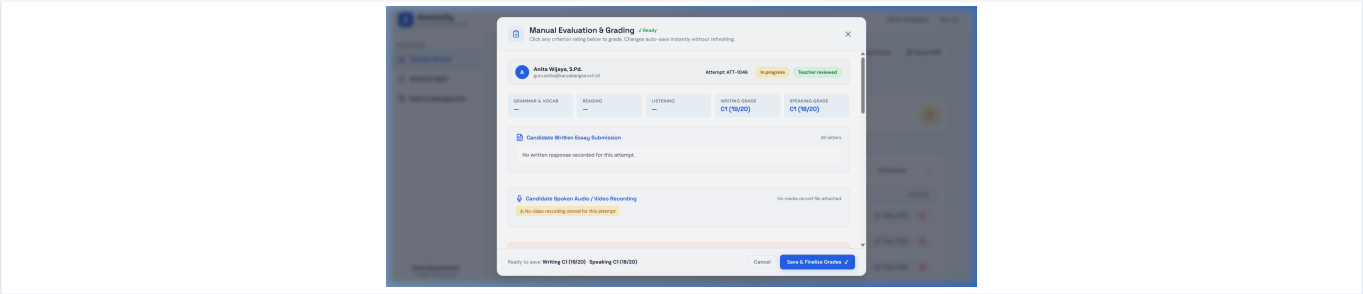
Dashboard Metrics & Filtering

- Quick stats cards: Total Candidates, Completed Tests, Pending Reviews.
- Instant school unit filter tabs: All Units, TK, SD, SMP, SMA, and SMK Karya Bangsa.
- Real-time search bar filtering across candidate names, emails, and attempt IDs.

Data Actions & Governance

- Export to styled Excel (.xlsx) with automated column widths and formulas.
- Export to centered PDF placement certificate for individual/all records.
- Batch selection checkboxes with safe confirmation modal for bulk deletion.

LIVE APPLICATION CAPTURE: Candidate Submission Details Modal (Objective Breakdown & Responses)



Candidate Details Inspector

- Detailed breakdown of raw scores vs. CEFR bands across all 5 skills.
- Full transcript view of typed writing essay and submitted objective responses.
- Speaking recording audio/video player with full scrubber controls.

Quality Assurance & Audit Trail

- Timestamped audit trail showing start time, completion time, and reviewer.
- Certificate serial number generation (e.g. KBS-EN-2026-1046).
- Instant access to candidate media stream without external software.

11. Administrative Rubric Grading Matrix

Assessify incorporates an interactive 4-criteria rubric grading modal for Writing and Speaking. Administrators evaluate qualitative performance on a 1–5 scale, with instant automated summation and CEFR band calculation.

LIVE APPLICATION CAPTURE: Interactive 4-Criteria Rubric Grading Modal (Writing & Speaking 1–5 Scale)

Writing Rubric Evaluation (4 Criteria)

1. Task Achievement (1–5 pts): Addresses prompt, thesis development.
2. Coherence & Cohesion (1–5 pts): Paragraphing, logical transitions.
3. Lexical Resource (1–5 pts): Vocabulary range, academic precision.
4. Grammatical Range & Accuracy (1–5 pts): Sentence complexity, control.

Speaking Rubric Evaluation (4 Criteria)

1. Fluency & Coherence (1–5 pts): Natural pacing, rhythm, hesitation.
2. Lexical Resource (1–5 pts): Spoken vocabulary, idiom appropriacy.
3. Grammatical Range & Accuracy (1–5 pts): Syntactic flexibility.
4. Communication & Pronunciation (1–5 pts): Intelligibility, clarity.

LIVE APPLICATION CAPTURE: Finalized Assessment Result with Updated C1 Placement & "Teacher Reviewed" Status

TEACHER CANDIDATE	SCHOOL UNIT	ATTEMPT ID	STATUS	OVERALL BAND	REVIEW STATUS
Anita Wijaya, S.Pd	SMP KARYA BANGSA	ATT-1888	In Progress		Teacher review
Dr. Hendra Gunawan, M.Pd	SMA KARYA BANGSA	ATT-1889	Completed	A1	Teacher review
Dr. Hendra Gunawan, M.Pd	SMA KARYA BANGSA	ATT-1890	Completed	A1	Teacher review
Dr. Hendra Gunawan, M.Pd	SMA KARYA BANGSA	ATT-1891	Completed	A1	Teacher review
Dr. Hendra Gunawan, M.Pd	SMA KARYA BANGSA	ATT-1892	Completed	A1	Review request

Scoring Engine Logic & Band Assignment

- Point Thresholds: 4–6 pts = A1 | 7–9 pts = A2 | 10–13 pts = B1 | 14–17 pts = B2 | 18–20 pts = C1.
- Live badge update: evaluator sees calculated CEFR band immediately upon button selection.
- "Save & Finalize Grades" locks the manual review and updates overall candidate placement.

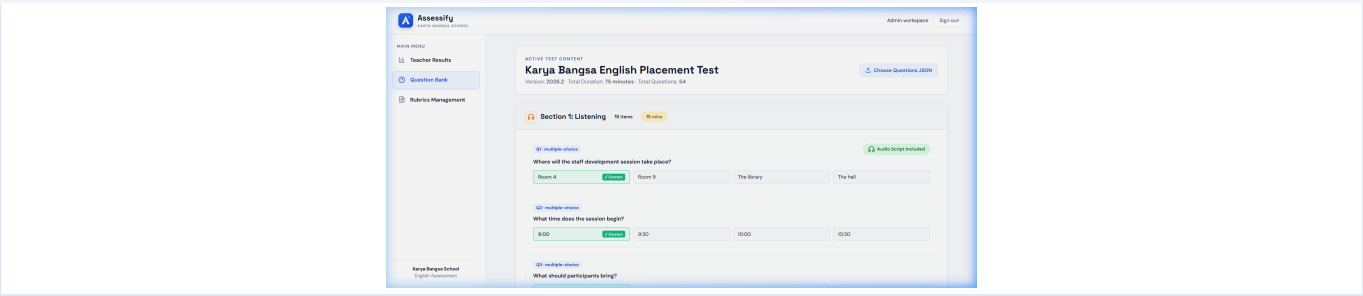
Review Status Lifecycle

- "Pending" -> In progress test.
- "Writing and Speaking review required" -> Submitted.
- "Review in progress" -> Partial criteria graded.
- "Teacher reviewed" -> Both Writing & Speaking fully evaluated.

12. Question Bank & Rubric Schema Management

Assessify provides native JSON schema management interfaces allowing academic coordinators to update question banks, reconfigure section timings, alter answer keys, and adjust rubric definitions without requiring code redeployment.

LIVE APPLICATION CAPTURE: Question Bank Management Interface (JSON Schema Editor, Upload & Live Preview)



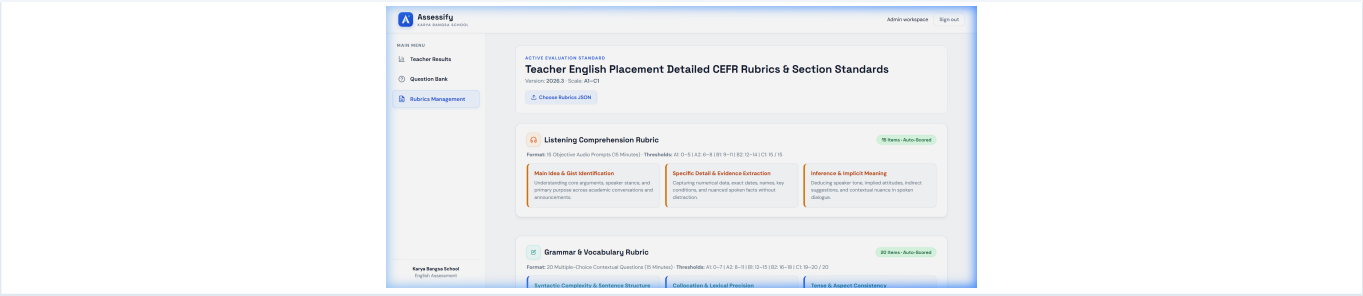
Question Bank Editor Capabilities

- Live formatted JSON viewer displaying active sections, questions, and keys.
- Client-side JSON schema validation preventing malformed syntax.
- Direct file upload (.json) and download of active question banks.
- Instant synchronization with server filesystem and cache.

Content Governance & Safety

- Automated sanitization removing correct answer keys before sending to clients (`safeTest()`).
- Support for multiple question types: single choice, reading passage, audio prompt, essay.
- Configurable section time allocations and instructions.

LIVE APPLICATION CAPTURE: Rubrics & Standards Management Interface (CEFR Criteria & Threshold Editor)



Rubrics Configuration Features

- Structured descriptors for all 5 skill areas aligned with official CEFR standards.
- Configurable score-to-band conversion thresholds for objective items.
- Downloadable institutional rubric template in JSON format for backup and versioning.

Institutional Standards Alignment

- Synchronized with official IELTS Band Descriptors (Public Version).
- Meets Indonesian Ministry of Education (Kemendikbudristek) teacher competency standards.
- Exportable audit reports for school accreditation across TK, SD, SMP, SMA, SMK.

13. Mathematical Scoring Engine & Composite Placement Model

The Assessify placement engine combines deterministic objective mapping with weighted criterion summation to assign overall CEFR levels.

Skill / Section	Items / Scale	Scoring Mode	B2 Threshold	C1 Threshold
Listening Comprehension	15 Items	Automated Answer Key	12–14 / 15 correct	15 / 15 correct
Grammar & Vocabulary	20 Items	Automated Answer Key	16–18 / 20 correct	19–20 / 20 correct
Reading Comprehension	15 Items	Automated Answer Key	12–14 / 15 correct	15 / 15 correct
Writing Assessment	4 Criteria (1–5)	Manual Admin Rubric	14–17 / 20 points	18–20 / 20 points
Speaking Assessment	4 Criteria (1–5)	Manual Admin Rubric	14–17 / 20 points	18–20 / 20 points

14. Composite Overall Placement Algorithm

- The composite overall band is computed via a multi-skill profile rule ensuring balance across receptive and productive skills:
- C1 Placement: At least 3 skills at C1, with Writing "e B2 and Speaking "e B2.
 - B2 Placement: At least 3 skills at B2 or C1, with Writing "e B1 and Speaking "e B1.
 - B1 Placement: At least 3 skills at B1 or higher, with Writing "e A2 and Speaking "e A2.
 - A2 / A1 Placement: Assigned when criteria for higher intermediate bands are not met.

15. Production Readiness & Future Roadmap

Verified Production Strengths	Recommended Future Enhancements
• Pure native Node.js ESM architecture with zero memory leaks. • Dual MySQL / In-memory storage with automated graceful degradation. • Automated unit test suite (100% pass) verifying scoring boundary conditions. • High-performance PDF & Excel export pipelines with certificate serials.	• Enterprise Single Sign-On (SSO / OIDC) integration. • Cloud object storage (AWS S3 / GCS) for multi-gigabyte media uploads. • Automated speech-to-text scoring assistance via AI acoustic models. • Anti-cheat tab-blur detection and browser fullscreen lockdown mode.